Universal AI Marketing Automation System (AIMAS)

Executive Summary

A modular, multi-agent AI system that autonomously builds and executes complete marketing strategies for any business vertical. Users provide minimal input (business type, target audience, budget, goals), and the system auto-discovers competitors, researches market, and orchestrates integrated campaigns across SEO, paid ads, cold outreach, sales, and customer support.

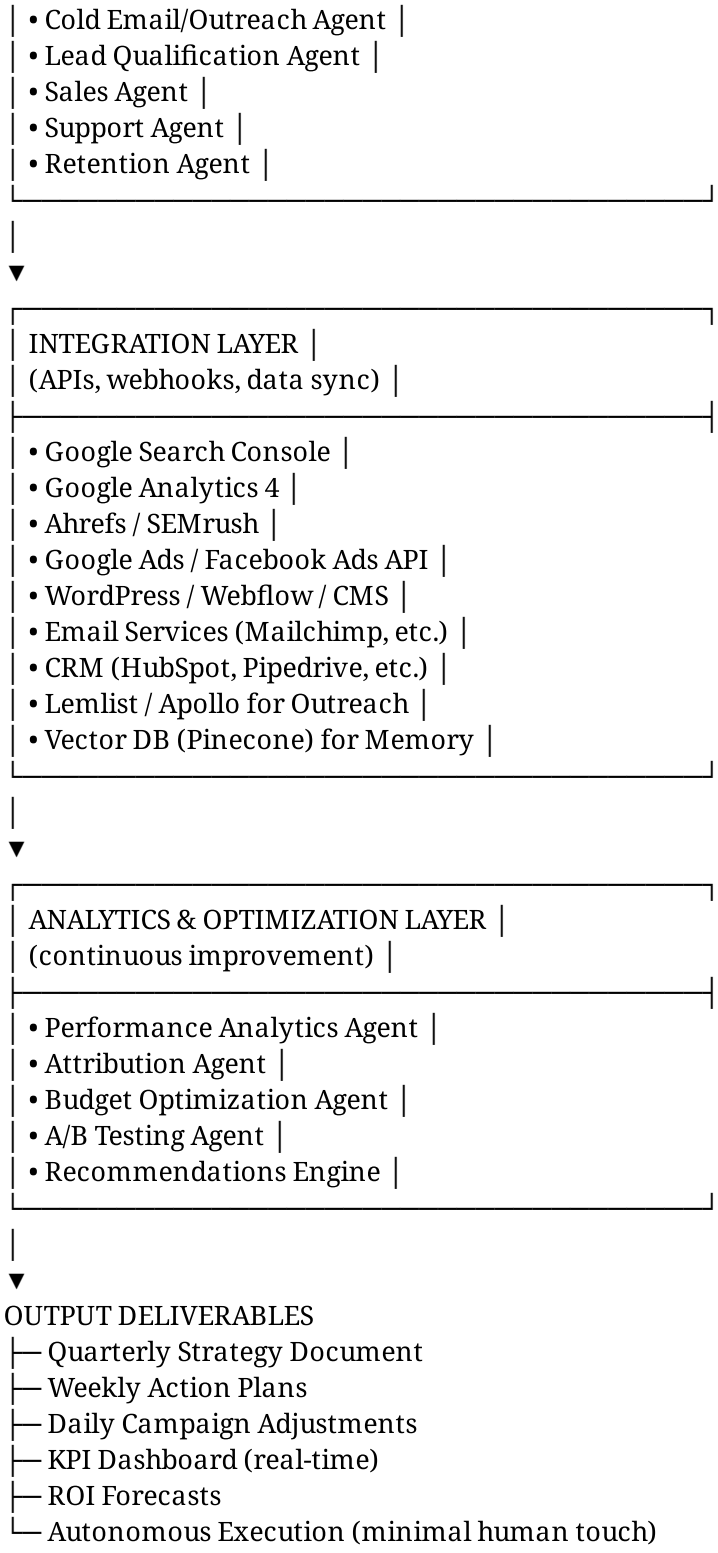
Core Promise: Full marketing automation from awareness to retention, adapting strategy by vertical without human intervention.

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1. System Architecture

1.1 High-Level Flow





1.2 Agent Hierarchy & Responsibilities

Tier 1: Discovery Agents (Research & Planning)

- **Market Research Agent**: Autonomous web search for market trends, TAM, emerging opportunities

- **Competitor Analysis Agent**: Find & analyze 5–10 main competitors (pricing, messaging, positioning, channels, traffic)
- **Audience Profiling Agent**: Build detailed personas (pain points, goals, preferred channels, search behavior)
- **Channel Selection Agent**: Based on audience & budget, recommend optimal marketing mix (% to SEO, PPC, Email, Outreach, etc.)
- **Budget Allocation Agent**: Distribute budget across channels based on expected ROI and customer lifetime value

Tier 2: Execution Agents (Do the Work)

- **SEO/Content Agent**: Keyword research → content calendar → on-page optimization → link strategy
- **Paid Ads Agent**: Campaign setup → audience targeting → copy variations → bid management
- **Outreach Agent**: Lead discovery → personalization → email sequences → follow-ups
- **Lead Qualification Agent**: Chatbot on website/email → scoring → handoff to sales
- **Sales Agent**: Autonomous outreach to qualified leads → objection handling → deal closing
- **Support Agent**: 24/7 customer support → issue resolution → escalation
- **Retention Agent**: Churn prediction → proactive outreach → upsell opportunities

Tier 3: Optimization Agents (Improve Continuously)

- **Analytics Agent**: Consolidate data from all sources → calculate CAC, LTV, ROAS, attribution
- **A/B Testing Agent**: Design tests → distribute traffic → analyze winners → recommend rollouts
- **Budget Optimizer**: Monitor performance → reallocate budget to winning channels
- **Insights Agent**: Find bottlenecks → suggest tactical changes → estimate impact

Tier 4: Orchestration & Memory

- **Master Orchestrator**: Routes tasks between agents, maintains conversation flow, preserves context
- **Long-Term Memory**: Vector DB stores competitor data, audience insights, previous campaign learnings

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2. Detailed Agent Specifications

2.1 Market Research Agent

Goal: Understand the market, trends, and opportunities for the business vertical

Inputs:

- Business type (e.g., "AI-powered project management SaaS")
- Target geography (e.g., "North America, EU, APAC")
- Target audience size (SMB / Enterprise)

Process:

1. Search: "market size SaaS project management 2025-2026"

2. Search: "emerging trends in collaboration tools"
3. Search: "[business type] adoption rates by company size"
4. Search: "regulatory changes affecting [industry]"
5. Compile insights: TAM, CAGR, growth drivers, headwinds

Outputs:

- Market overview (size, growth, trends)
- Audience segments and their buying power
- Seasonal patterns and peak demand periods
- Regulatory / compliance considerations

Tools Needed:

- Web search (Tavily, SerpAPI)
- LLM (Claude) for synthesis
- Vector storage for persistence

2.2 Competitor Analysis Agent

Goal: Map competitive landscape, identify gaps, and find positioning opportunity

Inputs:

- Business name / URL (or description)
- Target audience

Process:

1. **Company Discovery:**
   - Search: "[business type] competitors" → get list of 5–15 direct competitors
   - For each: scrape website, check Crunchbase, LinkedIn
2. **Detailed Analysis (for top 5):**
   - **Positioning:** How do they position? What's their core message?
   - **Pricing:** What are they charging? Pricing tiers, contract terms?
   - **Features:** What features do they highlight?
   - **Traffic & SEO:** Ahrefs data – domain authority, top keywords, estimated traffic
   - **Advertising:** What keywords are they bidding on? (SEMrush Ads)
   - **Brand Sentiment:** Reddit, Twitter, reviews – what do customers like/dislike?
   - **Channels:** Where are they active? (Twitter, LinkedIn, YouTube, etc.)
3. **Competitive Matrix:**
   - Features comparison table
   - Price comparison
   - Market positioning map (value vs. price)
4. **Opportunity Gap:**
   - What's underserved? (e.g., "No one is targeting mid-market with low-code option")
   - Where can we win?

Outputs:

- Competitor profiles (5 competitors, top strengths/weaknesses)
- Competitive positioning map

- Feature gaps and opportunities
- Recommended positioning angle
- Keyword opportunities (competitors rank for X, we don't)

Tools Needed:

- Ahrefs API (domain data, keywords, backlinks)
- SEMrush API (paid keywords, ads)
- Web scraping (BeautifulSoup / Selenium)
- Crunchbase / Hunter API

2.3 Audience Profiling Agent

Goal: Build detailed buyer personas and understand their decision journey

Inputs:

- Target audience description (e.g., "CTOs at mid-market SaaS companies")
- Business type
- Geographic focus

Process:

1. **Primary Research:**
    - Search Reddit, Twitter for "[target audience] pain points"
    - LinkedIn job searches to understand roles/titles
    - Industry reports on buyer priorities
2. **Persona Building:**
    - Pain points (top 5)
    - Goals and desired outcomes
    - Budget constraints
    - Decision-making process (who's involved? budget owner? technical evaluator?)
    - Preferred learning channels (blogs, YouTube, LinkedIn, podcasts)
    - Objections and concerns
3. **Search Intent Mapping:**
    - What do they search for at each stage of buying?
    - Awareness: "what is [problem]", "how to solve [problem]"
    - Consideration: "best tools for X", "[competitor] vs [competitor]"
    - Decision: "pricing comparison", "implementation guide"
4. **Create 3–5 Personas:**
    - Economic buyer (budget decisions)
    - Technical buyer (features & integration)
    - End-user (usability)

Outputs:

- Persona documents (name, role, goals, pain points)
- Buying journey map (touchpoints, information needs)
- Search intent by stage
- Messaging angles for each persona

Tools Needed:

- Web search
- Reddit/Twitter API
- LinkedIn scraping
- LLM for synthesis

2.4 Channel Selection Agent

Goal: Recommend optimal marketing channel mix based on audience and budget

Inputs:

- Audience segments and their preferences
- Monthly budget
- Business model (B2B, B2C, hybrid)
- Timeline (quick wins vs. long-term)
- Current assets (existing audience, brand, content)

Process:

1. **For each channel, calculate:**
    - Customer Acquisition Cost (CAC) based on industry benchmarks
    - Time to first conversion
    - Scalability potential
    - Fit with audience
2. **Channels to evaluate:**
    - **Organic Search (SEO):** Low CAC, 3–6 month ramp, highly scalable
    - **Paid Search (Google Ads):** Medium-high CAC, immediate, scalable, but burns budget fast
    - **Social Paid (Facebook, LinkedIn, TikTok):** Medium CAC, depends on platform and audience
    - **Email:** Ultra-low CAC if warm list, best for retention
    - **Cold Outreach:** Medium CAC, good for B2B, personalized
    - **Content/Affiliates:** Low CAC if structured right, takes time to set up
    - **Partnerships/PR:** Medium CAC, high-impact if structured right
3. **Recommend allocation:**
    - E.g., "SaaS to SMB: 40% SEO, 30% Google Ads, 20% LinkedIn, 10% Cold Email"
    - E.g., "E-commerce: 50% Google Shopping, 30% Facebook/Instagram, 15% Email, 5% Organic"

Outputs:

- Channel recommendation table
- Rationale for each channel
- Timeline (which channels to start first?)
- Budget allocation (%)

Tools Needed:

- Industry benchmark data (Gartner, HubSpot, Forrester reports)
- LLM reasoning

2.5 Budget Allocation Agent

Goal: Distribute monthly budget across channels and tactics for maximum ROI

Inputs:

- Monthly budget
- Channel mix (from Channel Selection Agent)
- Business goals (revenue target, lead volume, subscribers)
- LTV projections (estimated customer lifetime value)

Process:

1. **For each channel:**
    - Calculate expected customers per $1000 spent (based on benchmarks + competitor data)
    - Calculate CAC by channel
    - Calculate payback period
2. **Optimize for:**
    - Quick wins (ads, cold email) to fund long-term plays (SEO)
    - Risk distribution (don't put all budget in one channel)
    - Scalability (SEO and affiliate have higher scalability than PPC)
3. **Time allocation:**
    - Month 1–2: Focus on fast channels (ads, cold email) for revenue
    - Month 3–6: Build SEO foundation, email list
    - Month 6+: Scale what works

Outputs:

- Budget breakdown by channel ($)
- Budget breakdown by tactic (e.g., "$2000 for content creation, $1500 for link building")
- Expected monthly outcomes (leads, revenue, subscribers)
- Payback period
- Updated allocation every month based on performance

Tools Needed:

- LLM for optimization
- Spreadsheet generation (Airtable, Google Sheets API)

2.6 SEO/Content Agent

Goal: Build organic search presence through keyword targeting and content strategy

Inputs:

- Target keywords (from Competitor Analysis)
- Audience personas
- Business model and monetization
- Budget
- Existing content

Process:

Phase 1: Keyword Research & Strategy

1. Query Ahrefs API:
    - Get search volume, keyword difficulty for 100–200 keywords
    - Analyze SERP: "who ranks?" (analyze top 10 results)
2. Cluster keywords by:
    - Intent (informational, commercial, transactional)
    - Buyer stage (awareness, consideration, decision)
    - Topic (group related keywords)
3. Prioritize:
    - Low difficulty + decent volume = quick wins
    - High volume + medium difficulty = long-term plays
    - Long-tail keywords = quick to rank, immediate traffic
4. Create content map:
    - Pillar pages (broad topics, cluster hubs)
    - Cluster content (supporting pages linked to pillars)
    - Bottom-funnel content (conversion-focused)

Phase 2: Content Calendar

1. Generate 12-month calendar:
    - Week 1–4: Foundation content (basics, terminology, guides)
    - Month 2–3: Middle funnel (comparisons, case studies, how-tos)
    - Month 4–6: Authority content (research, benchmarks, industry reports)
    - Ongoing: Trend-jacking, user questions, seasonal content
2. For each article:
    - Primary keyword + secondary keywords
    - Target audience persona
    - Estimated traffic potential
    - Content type (blog, video, infographic, interactive tool)
    - Monetization angle (CTA, affiliate link, product pitch)

Phase 3: Content Creation

1. Use Claude to generate outlines and first drafts
2. Optimize for:
    - Readability (short paragraphs, headers, lists)
    - SEO (keyword placement, meta tags, schema markup)
    - Conversion (CTAs, forms, links to product)
    - User experience (internal links, visual breaks)
3. Publish workflow:
    - Auto-post to WordPress via API
    - Auto-generate meta descriptions
    - Auto-set internal links

Phase 4: Link Building & Authority

1. Find link opportunities:
    - Competitor backlink analysis (Ahrefs)
    - Broken link opportunities
    - Resource pages

    - Industry directories
2. Outreach:
    - Use cold email agent to pitch guest posts
    - Reach out for resource page inclusion
    - PR / partnerships for brand mentions
3. On-page optimization:
    - Update old content with new data
    - Internal linking (cross-link cluster content)
    - Improve metadata

Outputs:

- Keyword strategy (100–200 keywords ranked by priority)
- 12-month content calendar
- Content templates (blog post, case study, comparison)
- Monthly reporting (rankings, traffic, conversions)
- Recommendations for next month

Tools Needed:

- Ahrefs API (keyword research, SERP analysis, backlink data)
- Claude API (content generation)
- WordPress API (publishing)
- GA4 API (traffic tracking)
- Tavily (news/trend discovery)

2.7 Paid Ads Agent

Goal: Run profitable paid ad campaigns across Google Ads, Facebook, LinkedIn

Inputs:

- Budget
- Target audience (from Audience Profiling Agent)
- Competitor keywords (from Competitor Analysis)
- Landing page / offer
- Goals (leads, sales, brand awareness)

Process:

Campaign Strategy:

1. **Google Search Ads:**
    - Bid on high-intent keywords (bottom-funnel)
    - Competitors' brand keywords (e.g., "Notion alternative")
    - "Best [solution]" keywords
2. **Facebook / Instagram:**
    - Interest-based targeting (lookalike audiences)
    - Behavioral targeting (website visitors, email subscribers)
    - Demographic targeting (job title, income, interests)
3. **LinkedIn (for B2B):**
    - Job title targeting
    - Company size, industry

    - Seniority level

Campaign Setup (Autonomous):

1. Create ads automatically via API:
    - Generate 5 headline variations
    - Generate 3 body copy variations
    - Design creative (using DALL-E or template library)
2. Audience segmentation:
    - Cold (awareness) → brand messaging
    - Warm (website visitors) → consideration messaging
    - Hot (email subscribers) → product/pricing messaging
3. Bidding strategy:
    - Google Ads: Target CPA bid strategy
    - Facebook: Lowest cost delivery initially, then optimize for ROAS
4. Landing pages:
    - Auto-generate variants
    - Optimize copy, CTA, form fields

Optimization Loop (Daily):

1. Check performance:
    - CTR, CPC, CAC, conversion rate
    - Compare to targets
2. Pause underperformers
3. Increase budget for winners
4. Test new ad copy/creative
5. Adjust bids

Outputs:

- Campaign performance dashboard (daily)
- Weekly optimization recommendations
- Monthly reporting (ROAS, CAC, LTV, ROI)
- Budget allocation adjustments

Tools Needed:

- Google Ads API
- Facebook Ads API
- LinkedIn Campaign Manager API
- DALL-E or stable diffusion (ad creative)
- LLM (copy generation)

2.8 Outreach Agent (Cold Email)

Goal: Identify and qualify B2B leads through personalized outreach

Inputs:

- Target audience (titles, companies, geographies)
- Budget for outreach tools
- Business model and pitch

Process:

Lead Discovery:

1. Query Apollo.io or Hunter.io API for prospects:
    - Job title + company filters
    - Industry filters
    - Company size filters
    - Geography
2. Enrich data:
    - Get emails, phone numbers
    - Pull LinkedIn profiles
    - Find decision makers

Personalization:

1. For each lead:
    - Research company (web search, LinkedIn Company page)
    - Identify pain points (based on company type + industry trends)
    - Find personal connection (went to same school, shared connections, etc.)
2. Generate personalized email:
    - Hook: reference something specific to them
    - Problem: articulate their pain
    - Solution: how your product solves it
    - CTA: clear next step (demo, call, trial)

Sequencing:

1. Email sequence (3–5 emails):
    - Day 0: Initial cold email
    - Day 3: Follow-up (if no reply)
    - Day 7: Different angle
    - Day 14: Last attempt
    - Day 21: Remove from sequence
2. Timing optimization:
    - Send Tuesday–Thursday, 9–11 AM recipient timezone
    - Avoid weekends and holidays

Tools:

- Lemlist or Instantly API (email sending + tracking)
- Apollo.io API (lead discovery)
- Hunter.io API (email finding)
- LLM (personalization)

Outputs:

- Weekly campaign performance
    - Emails sent, opened, replied
    - Reply rate, booking rate
    - Cost per booked meeting
- Lead quality (MQL score)
- A/B test results (subject lines, body copy)

2.9 Lead Qualification Agent (Chatbot)

Goal: Qualify leads in real-time, capture intent, and hand off to sales

Inputs:

- Website URL
- Target audience personas
- Product/offer
- Qualifying criteria (budget, company size, timeline, need)

Process:

Chatbot Deployment:

1. Embed on website (Intercom, Drift, Chatbase)
2. Trigger on:
    - High-intent page (pricing page, comparison page)
    - Scroll depth > 50%
    - Exit intent

Qualification Flow:

1. Greeting:
    - "Hi! Are you looking to [solve problem X]?"
2. BANT questions:
    - **Budget:** "What's your approx. budget for this?" (Yes/No thresholds)
    - **Authority:** "Are you involved in purchasing decisions?" (Decision maker?)
    - **Need:** "How urgent is solving this?" (timeline)
    - **Timeline:** "When are you looking to implement?" (3 months? 6 months? ASAP?)
3. Scoring:
    - Assign points for each answer
    - Total: 0–100 score
    - **Hot (80+):** Trigger immediate sales agent contact
    - **Warm (50–79):** Add to nurture sequence
    - **Cold (<50):** Continue nurturing content
4. Data capture:
    - Email, name, company
    - Problem summary (user's own words)
    - Next step (schedule demo, send info, etc.)

Integration:

- Hot leads → CRM, trigger sales agent email
- All data → HubSpot or Pipedrive CRM

Outputs:

- Real-time lead quality dashboard
- Weekly conversion rates (visitor → qualified lead)
- Chat transcript analysis (common objections, questions)
- Recommendations to improve messaging

Tools Needed:

- Intercom / Drift / Chatbase (platform)
- Claude API (conversation generation)
- Zapier / Webhook (CRM sync)

2.10 Sales Agent

Goal: Autonomously close deals with qualified leads

Inputs:

- Qualified lead profile (from Lead Qualification Agent)
- Product details, pricing, positioning
- Historical sales playbooks (objection handling, negotiation ranges)
- CRM integration

Process:

Lead Engagement:

1. Receive hot lead from chatbot
2. Within 5 minutes: Send personalized email
    - Reference conversation
    - Offer: "I see you're interested in X. Happy to do a 20-min call this week?"
    - Calendar link (Calendly)

Demo / Call:

1. If they schedule: Prepare context
    - Pull their company info, industry, use case
    - Pre-answer likely questions
2. During call: Agent (human-in-the-loop or fully autonomous for simple cases):
    - Confirm problem
    - Show how product solves it
    - Handle objections
    - Propose terms

Negotiation:

1. If customer hesitates on price:
    - Auto-generate discount offer (if within approval limits)
    - E.g., "20% off if you commit to annual" or "30% off for first 3 months"
    - Terms: can adjust based on deal size
2. If customer wants features not included:
    - Offer custom pricing or upgrade path
3. Close:
    - Send contract or payment link
    - Follow-up reminders if unsigned

Automation Level:

- **Low touch:** Email sequences + Calendly scheduling, human closes
- **Mid touch:** Chatbot qualifies → automated emails + human call

- **High touch:** AI fully autonomous for small deals (<$500/mo), human for large deals

Outputs:

- Daily sales pipeline (leads, stage, expected close date)
- Deal win rate and average deal size
- Sales velocity (time from first touch to signature)
- Objection analysis and handling recommendations

Tools Needed:

- Calendly API (booking)
- Email API (SendGrid, Mailchimp)
- CRM API (HubSpot, Pipedrive)
- Stripe / payment API (invoicing)
- DocuSign API (e-signature)
- LLM (email copy, negotiation logic)

2.11 Support & Service Agent

Goal: Provide 24/7 customer support, resolve issues, reduce churn

Inputs:

- Product knowledge base
- Common issues / FAQs
- Customer history (from CRM)
- Product usage data (if SaaS)

Process:

Issue Resolution:

1. Customer submits ticket or opens chat
2. Agent analyzes:
    - Issue type (technical, billing, feature request)
    - Customer value (are they at risk of churn?)
    - Urgency (critical vs. low priority)
3. Resolution path:
    - Tier 1 (70%): Can AI solve? (password reset, FAQ, troubleshooting)
    - Tier 2 (20%): Needs human but AI can prep context and possible solutions
    - Tier 3 (10%): Escalate immediately (critical issue, VIP customer)
4. For Tier 1 issues:
    - Provide step-by-step solution
    - Ask for confirmation of resolution
    - Document for future reference
5. For Tier 2/3:
    - Prepare detailed ticket for human agent
    - Include: customer history, previous issues, usage data, revenue impact
    - Assign to appropriate specialist

Proactive Support:

- Monitor product usage: if customer's usage drops → proactive email
  - "Hi, noticed you haven't used feature X. It can help you Y. Can I help?"
- Churn risk detection:
  - Haven't logged in for 14 days? → Reach out
  - Low feature adoption? → Onboarding email
  - Support tickets increasing? → Account health risk

Outputs:

- Daily support metrics (tickets, avg resolution time, CSAT score)
- Churn risk list (prioritize retention outreach)
- Knowledge base updates (common issues → FAQ)
- Customer health scores

Tools Needed:

- Intercom / Zendesk / Freshdesk API (ticketing)
- Product analytics API (usage tracking)
- Slack (for escalation)
- Email API
- LLM (response generation)

2.12 Retention & Upsell Agent

Goal: Maximize customer lifetime value, prevent churn, identify upsell opportunities

Inputs:

- Customer data (spending, usage, contract end date)
- Product portfolio and upsell paths
- Retention budgets (how much can we discount to save a customer?)

Process:

Churn Prevention:

1. Identify at-risk customers:
   - Haven't used product in 30 days
   - Support tickets increasing (unhappy signal)
   - Contract ending soon without renewal intent
   - Spending decreased month-over-month
2. Intervention:
   - **Engaged users:** Send feature tips, best practices, case studies
   - **Low engagement:** "We notice you haven't used X. Let's jump on a call to maximize your ROI"
   - **Churning:** Special offer (discount, additional features, extended trial)
3. Proactive outreach:
   - 60 days before contract end: "Renewing soon? Let's review your usage and see if we can increase value"
   - Offer: upgrade, bundle, annual discount, etc.

Upsell Opportunities:

1. Analyze customer:
    - Current plan tier
    - Feature requests they've made
    - Growth in usage (more users, more data)
2. Recommend:
    - "Your team has grown to 15 users. Pro plan might give you admin controls"
    - "You're using integration extensively. Enterprise plan includes dedicated support"
    - "Add-on: advanced reporting ($200/mo) could save you 5 hours/week"
3. Campaign:
    - Email sequence (3 emails)
    - Personalized demo of new features
    - Limited-time upgrade discount

Outputs:

- Churn risk dashboard (customers at risk, by severity)
- Renewal pipeline (contracts coming up)
- Upsell pipeline (opportunities, value, probability)
- Monthly: retention rate, upsell rate, NRR (Net Revenue Retention)

Tools Needed:

- CRM API (customer data)
- Product analytics API (usage)
- Email API
- LLM (personalized messaging)

2.13 Analytics & Intelligence Agent

Goal: Consolidate data from all sources, calculate metrics, provide insights and recommendations

Inputs:

- Data from all agents and tools (GA4, SEO tools, ads, email, CRM, product)
- Business goals and targets
- Previous month's results

Process:

Data Collection (Daily):

1. Pull from all sources:
    - Google Analytics 4: traffic, conversions, user behavior
    - Google Search Console: impressions, clicks, rankings
    - Ahrefs: organic rankings, traffic estimates
    - Google Ads: spend, clicks, conversions
    - Facebook Ads: same metrics
    - Email: opens, clicks, unsubscribes
    - Lemlist: email opens, replies, bookings
    - CRM: deals, pipeline, win rate
    - Product: signups, active users, churn

    - Revenue: MRR, ARR, refunds
2. Normalize and store:
    - Append date, source, channel
    - Store in data warehouse (Airtable, Google Sheets, Supabase)

Metrics Calculation (Weekly):

1. **Acquisition metrics:**
    - CAC (Cost per Acquired Customer) by channel
    - CAC payback period (how long to recoup acquisition cost)
    - Traffic by source
    - Conversion rate by channel
    - Lead volume and quality
2. **Monetization metrics:**
    - Revenue by source
    - Average deal size
    - Sales cycle length
    - Win rate
3. **Retention metrics:**
    - Churn rate
    - NRR (Net Revenue Retention)
    - Upsell rate
    - Customer satisfaction (CSAT, NPS)
4. **Efficiency metrics:**
    - CAC:LTV ratio (should be at least 1:3)
    - Marketing efficiency ratio (revenue per marketing dollar)
    - Payback period
5. **Attribution:**
    - Which touchpoints led to conversion?
    - Multi-touch attribution (if user saw 3 ads, 1 email, 1 blog article before buying)
    - Last-click attribution (which channel got credit for final conversion)

Insights & Recommendations (Weekly/Monthly):

1. Identify winners:
    - "Google Search Ads CAC is $80, LTV is $2400. Increase budget by 50%"
    - "Cold email reply rate 8%, booking rate 3%. Scale outreach"
2. Identify losers:
    - "Facebook CAC $300, LTV $1500, payback 6 months. Reduce budget or optimize"
    - "Email open rate dropped 15%. Consider new subject lines or list health issues"
3. Forecasts:
    - "If we increase SEO budget $500/mo, expect +2000 visits/mo in 3 months (based on scaling)"
    - "If we maintain current spend, MRR will reach $50k in 6 months. Need to accelerate? Recommend 30% budget increase."
4. Actionable recommendations:
    - "Top performers this week: SEO ($0 cost, $5k MRR pipeline), Cold Email ($2 CAC, booking rate 4%). Allocate 60% of new budget here."
    - "Opportunities: Facebook ROAS dropped. Test new audiences or creative. LinkedIn still cold. Skip for now."

Outputs:

- Weekly dashboard (KPIs, trends, status)
- Monthly performance report (what worked, what didn't, forecast)
- Budget optimization recommendation (reallocate based on performance)
- A/B test results and recommendations
- Cohort analysis (which cohorts are most valuable?)

Tools Needed:

- GA4 API
- Google Ads API
- Facebook Ads API
- Ahrefs API
- HubSpot API
- Product analytics API
- Google Sheets API (or Airtable)
- LLM (insights synthesis)
- Visualization library (Plotly, Matplotlib) for charts

2.14 A/B Testing Agent

Goal: Design, run, and analyze experiments to optimize conversion rates

Inputs:

- Current conversion rate by channel
- Elements to test (landing page, email subject, ad copy, CTA, design)
- Statistical power needed (95% confidence, 80% power)

Process:

Experiment Design:

1. Define:
    - **Hypothesis:** "Changing CTA from 'Free trial' to 'Get started' will increase CTR by 15%"
    - **Control:** Original version
    - **Variant:** New version
    - **Success metric:** Click rate, conversion rate, revenue
    - **Sample size:** How many visitors needed? (calculated based on baseline rate + desired lift + power)
    - **Duration:** How long to run? (until sample size reached or 2–4 weeks)
2. Design variants (auto-generate):
    - Landing page: Change headline, CTA, form fields, images
    - Email: Change subject line, sender name, opening, CTA
    - Ads: Change image, headline, description, CTA
3. Setup (via APIs):
    - Google Optimize (for website tests)
    - Email platform native testing
    - Ad platform native testing

Monitoring (Daily):

1. Check progress:

    - Sample size collected?
    - Interim results (directionally, which is winning?)
    - Stopping rule: if one variant is 95% confident winner, stop early and roll out

Analysis (At End of Test):

1. Calculate:
    - Conversion rate for each variant
    - Statistical significance (is difference real or random?)
    - Confidence interval
2. Recommend:
    - If variant wins: "New CTA increased conversion rate 12%, which is 95% confident. Recommend rolling out to 100% of traffic."
    - If control wins: "New CTA didn't improve results. Keep current version, try different approach."
    - If no winner: "No statistically significant difference. Try different test."
3. Learnings:
    - Document what works (use in future tests)
    - Update playbooks (sales, email, ads teams)

Outputs:

- Experiment backlog (next tests to run)
- Running experiments dashboard
- Completed experiments results
- Learning repository (what we know works)

Tools Needed:

- Google Optimize API
- Email platform API
- Google Analytics / GA4 API
- Statistical calculator (scipy, numpy in Python)
- LLM (hypothesis generation, learnings synthesis)

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3. Data Flows & Integrations

3.1 Core APIs & Integrations

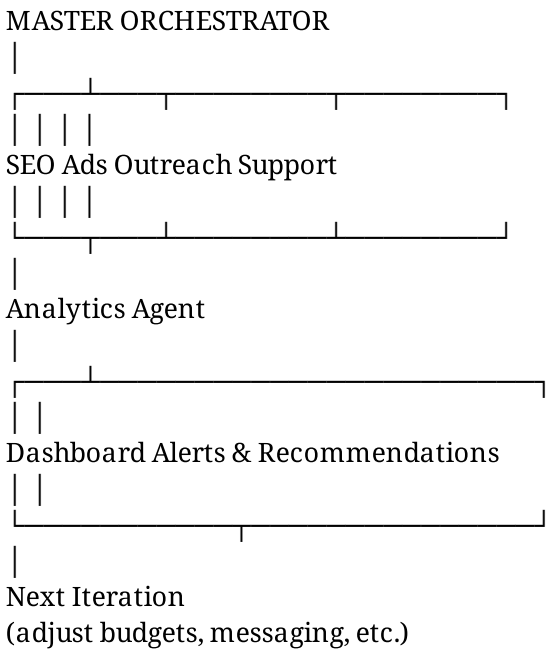
Category	Tool	API Available	Purpose
Search & Analytics	Google Search Console	✅	Organic rankings, clicks, impressions
	Google Analytics 4	✅	Traffic, user behavior, conversions
	Ahrefs	✅	Keyword research, backlinks, competitor analysis
	SEMrush	✅	PPC keywords, competitor ads
Paid Advertising	Google Ads	✅	Search, Shopping, Display ads
	Facebook Ads	✅	Feed, Stories, Reels ads
	LinkedIn Campaign Manager	✅	Sponsored content, messaging
Email & Outreach	Mailchimp	✅	Email campaigns, automation
	Sendgrid	✅	Transactional email
	Lemlist	✅	Cold email tracking
	Apollo.io	✅	Lead database, enrichment
	Hunter.io	✅	Email finding
CRM & Pipelines	HubSpot	✅	Customer data, deals, workflows
	Pipedrive	✅	Sales pipeline
Content Management	WordPress	✅	Blog publishing

Category	Tool	API Available	Purpose
	Webflow	✅	Website management
Payments & Billing	Stripe	✅	Invoicing, subscriptions
Web Scraping	Bright Data	✅	Competitor data scraping
Chat & Support	Intercom	✅	Chatbot, support tickets
	Freshdesk	✅	Support ticketing
Data Storage	Pinecone	✅	Vector database (memory)
	Airtable	✅	Structured data storage
	Supabase	✅	PostgreSQL + API layer

3.2 Data Sync Flow

Every 4 Hours:
GA4 → Pull traffic, conversions → Store in Data Lake
GSC → Pull rankings, clicks → Store in Data Lake
CRM → Pull deals, customers → Store in Data Lake
Email → Pull opens, clicks → Store in Data Lake
Ads → Pull spend, conversions → Store in Data Lake
Product → Pull signups, usage → Store in Data Lake
│
▼
Data Normalization
(same format for all sources)
│
▼
Analytics Agent
(calculate metrics, detect anomalies)
│
├─ Metrics Dashboard (real-time)
├─ Alerts (if metrics degrade)
└─ Recommendations (to other agents)

3.3 Agent Communication Flow



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4. User Interface & Inputs

4.1 Onboarding Flow

Step 1: Business Setup
□ Business Name: [text field]
□ Business Type: [dropdown: SaaS, E-commerce, Agency, Local, Creator, B2B Services, Other]
□ Website URL: [text field]
□ Current Monthly Revenue: [dropdown: <$5k, $5–50k, $50–500k, $500k+]
□ Target Monthly Revenue: [text field]
□ Timeline: [dropdown: 3 months, 6 months, 12 months]

Step 2: Target Audience
□ Who is your ideal customer? [long text]
Example: "CTOs at Series A SaaS companies, $2–10M ARR, in North America"

□ What problem do you solve? [long text]

□ How will they benefit? [long text]

Step 3: Budget & Goals
□ Monthly Marketing Budget: $[text field]
□ Primary Goal: [dropdown: Lead generation, Revenue, Brand awareness, Subscribers]
□ Secondary Goals: [checkboxes: Email list growth, Demo bookings, Product sales, Partnerships]

Step 4: Current Assets
□ Do you have a website? [Yes/No]
If Yes: Provide URL

□ Do you have existing content? [Yes/No]
If Yes: How much? [dropdown: <10, 10–50, 50–100, 100+]

□ Do you have an email list? [Yes/No]
If Yes: Size? [dropdown: <100, 100–1k, 1k–10k, 10k+]

□ Are you running ads? [Yes/No]
If Yes: Which platforms? [checkboxes: Google Ads, Facebook, LinkedIn]

Step 5: Constraints & Preferences
□ Industries/topics to avoid: [text field]
□ Channels to avoid: [checkboxes: Paid ads, Cold outreach, etc.]
□ Brand voice: [dropdown: Serious & formal, Friendly & conversational, Technical & expert, Playful & casual]
□ Preferred content types: [checkboxes: Blog, Video, Case studies, Webinars, Infographics]

4.2 Main Dashboard

Real-time KPI Summary:

```
┌──────────────────────────────────────────────────────────────────────────┐
│ Monthly MRR: $15,200 │ MoM Growth: +23% │ Projected ARR: $182,400 │
├──────────────────────────────────────────────────────────────────────────┤
│ CAC: $120 │ LTV: $3,600 │ Payback: 1.2 mo │ CAC:LTV: 1:30 │
└──────────────────────────────────────────────────────────────────────────┘
```

Channel Performance (Weekly):
Channel Revenue Leads CAC ROAS Trend Action
Organic (SEO) $5,200 18 $50 N/A ↑ +15% Scale
Google Ads $3,800 42 $90 2.5x → Flat Test copy
Cold Email $4,200 35 $80 N/A ↑ +8% Scale
Facebook Ads $2,000 20 $120 1.8x ↓ -12% Optimize

Marketing Funnel (Current Month):
Visitors Leads MQL SQL Customer Revenue

```
┌──────────┐ ┌──────────┐ ┌──────────┐ ┌──────────┐ ┌──────────┐ ┌──────────┐
│ 18,500 │→ │ 1,250 │→ │ 850 │→ │ 380 │→ │ 95 │→ │ $15,200 │
│ 100% │ │ 6.8% │ │ 68% │ │ 44.7% │ │ 25% │ │ 100% │
└──────────┘ └──────────┘ └──────────┘ └──────────┘ └──────────┘ └──────────┘
```

Agent Status:
✓ SEO Agent - 42 articles published, 156 keywords ranking
✓ Ads Agent - 8 campaigns running, $1,200/day spend
✓ Outreach Agent - 320 prospects, 8% reply rate, 3% booking rate

✅ Support Agent - 47 tickets resolved, avg resolution time 2.3 hours
 Sales Agent - 12 active deals, expected close: $45k next week
 Retention Agent - 3 at-risk customers identified, outreach scheduled

---

4.3 Weekly Recommendations Panel

```
┌── WEEK 8 RECOMMENDATIONS ──────────────────────────────────────────────┐
│                                                                        │
│  TOP OPPORTUNITIES                                                    │
│  ├─ Organic search trending up (+15%). Increase content               │
│  │  creation budget 30% ($300 → $390/week)                            │
│  │  Expected impact: +240 visitors/month (+12% MRR)                   │
│  │                                                                    │
│  ├─ Cold email booking rate 3%, above industry avg (2%).              │
│  │  Scale outreach 50% (320 → 480 prospects/week)                     │
│  │  Expected impact: +35 leads/month                                  │
│  │                                                                    │
│  └─ Facebook ROAS flat. Try new creative (carousel ads)               │
│  and lookalike audiences. Budget pause until optimized                │
│                                                                        │
│  AT-RISK METRICS                                                      │
│  ├─ Email open rate: 18% (down from 22% last month)                   │
│  │  Recommendation: Test new subject line hooks                       │
│  │  Run A/B test: Current vs. "How we..." format                      │
│  │                                                                    │
│  └─ Customer churn: 1 customer at risk ($1.2k MRR)                    │
│  Reason: No product usage in 14 days                                  │
│  Action: Proactive onboarding call scheduled                          │
│                                                                        │
│  FORECAST (NEXT 30 DAYS)                                              │
│  ├─ MRR projected: $17,100 (+12% vs this month)                       │
│  ├─ Expected revenue per $1 marketing spend: $8.40                    │
│  ├─ CAC trend: Decreasing (good, indicates efficiency)                │
│  └─ Payback period: Improving (1.5mo → 1.2mo)                         │
│                                                                        │
│  ✅ NEXT ACTIONS (APPROVE TO AUTO-EXECUTE)                            │
│  ├─ [ ] Increase SEO content budget to $390/week                      │
│  ├─ [ ] Scale cold email: 320 → 480 prospects/week                    │
│  ├─ [ ] Pause Facebook ads pending creative test                      │
│  ├─ [ ] Launch email subject line A/B test                            │
│  └─ [ ] Schedule customer churn prevention call                       │
│                                                                        │
└────────────────────────────────────────────────────────────────────────┘
```

4.4 Content Calendar & Execution Board

SEO CONTENT CALENDAR (Next 12 Weeks)

Week Priority Keyword Intent Type Status Traffic Potential
---

1 High "How to automate Info Blog Writing 200 visits/mo
marketing"

2 High "Best marketing Comp Compare Outline 150 visits/mo
automation tools"

3 Medium "AI for SEO 2026" Info Guide Queued 120 visits/mo

4 High "[Product] vs Comp Case Writing 180 visits/mo
[Competitor]" Study

5 Low "Marketing trends Info Report Queued 90 visits/mo
2026"

...

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5. Configuration & Customization

5.1 Vertical-Specific Presets

System includes pre-configured setups for common verticals:

SaaS to B2B SMB:

- Channel mix: 40% SEO, 30% Google Ads, 20% Cold Email, 10% Content/Affiliates
- Content focus: How-to, comparisons, ROI calculators
- Sales: Lead magnet → email nurture → sales call
- Typical CAC: $80–200, LTV: $2,400–8,000

E-commerce:

- Channel mix: 50% Google Shopping, 30% Facebook/Instagram, 15% Email, 5% Organic
- Content focus: Product reviews, gift guides, tutorials
- Sales: Browse → cart → checkout, retargeting for abandonment
- Typical CAC: $5–50, LTV: $200–1,000

B2B Services (Agency, Consulting):

- Channel mix: 35% Content, 25% LinkedIn, 20% Cold Outreach, 20% Referrals
- Content focus: Case studies, thought leadership, benchmarks
- Sales: Content → lead magnet → qualification call → proposal
- Typical CAC: $500–3,000, LTV: $20,000–200,000

Creator / Info Products:

- Channel mix: 40% Content, 30% Email, 15% Affiliates, 15% Paid Social

- Content focus: Tutorials, personal stories, exclusive content
- Sales: Free content → email list → course/membership
- Typical CAC: $5–20, LTV: $100–1,000

Local Business:

- Channel mix: 50% Local SEO, 30% Google Local Ads, 15% Social, 5% Affiliates
- Content focus: Local reviews, service pages, local news
- Sales: Directory → calls, bookings, walk-ins
- Typical CAC: $20–100, LTV: $1,000–5,000

5.2 Constraints & Safety Rails

Budget Constraints:

- Set daily spend limits per channel
- System alerts if projection exceeds monthly budget
- Can pause/reduce channels if CAC exceeds threshold

Brand Safety:

- Specify industries/topics to avoid (e.g., crypto, gambling, political)
- System screens all content before publishing
- Content must pass brand voice check (formal vs. casual, etc.)

Competitive Safeguards:

- Monitor for price wars (if competitor undercuts, system flags)
- Don't outbid on competitor brand keywords (too expensive)
- Avoid direct negative comparisons (legal risk)

Privacy & Compliance:

- GDPR: consent collection for emails, data deletion on request
- CCPA: privacy policy, data handling
- CAN-SPAM: unsubscribe links, sender identification
- Industry-specific (e.g., HIPAA for health, SOC2 for SaaS)

---

6. Implementation Roadmap

Phase 1: MVP (Weeks 1–4)

What ships:

- Market research agent (auto-analyze industry)
- Competitor analysis agent (find + analyze 5 competitors)
- Audience profiling agent (build personas)
- Channel selection agent (recommend mix)
- Budget allocation agent (distribute spend)
- Basic analytics dashboard

Technology:

- Claude 3.5 Sonnet for reasoning
- CrewAI for agent orchestration
- Airtable for data storage
- Google Sheets API for spreadsheet output

Inputs required:

- Business type, target audience, budget, goals

Outputs:

- Quarterly strategy document (20–30 pages)
- Recommended channel mix and budget allocation
- Competitor analysis
- Audience personas

Phase 2: Execution Agents (Weeks 5–8)

What ships:

- SEO/Content Agent (keyword research, content calendar, publishing automation)
- Paid Ads Agent (Google Ads, Facebook setup and management)
- Outreach Agent (lead discovery, cold email sequencing)
- Lead Qualification Chatbot (embed on website)

Technology:

- WordPress API integration
- Google Ads API
- Facebook Ads API
- Lemlist API
- Intercom / Drift chatbot embedding

New inputs:

- Website URL
- Existing content/assets

New outputs:

- Content calendar (12 months)
- Ad campaigns (copy, creative, audience)
- Lead list + email sequences
- Chatbot script and embed code

Phase 3: Revenue Agents (Weeks 9–12)

What ships:

- Sales Agent (autonomous outreach, deal closing)
- Support Agent (customer support ticketing)
- Retention Agent (churn prevention, upsell)

Technology:

- CRM API (HubSpot, Pipedrive)
- Email automation
- Product analytics integration
- Stripe API

New outputs:

- Sales pipeline and forecast
- Support ticket automation
- Churn risk alerts
- Upsell opportunities

Phase 4: Optimization & Intelligence (Weeks 13+)

What ships:

- Analytics Agent (consolidated dashboards, KPI tracking)
- A/B Testing Agent (experiment design and analysis)
- Budget Optimizer (continuous reallocation)
- Insights Engine (recommendations)

Technology:

- GA4 API
- Statistical analysis (scipy, numpy)
- Data visualization (Plotly)
- Real-time dashboards

New outputs:

- Weekly performance reports
- Actionable recommendations
- Budget reallocation suggestions
- Experiment results and learnings

---

7. Pricing & Monetization

Option 1: Tiered SaaS

Tier 1 (Strategy Only): $299/month

- Strategy agent (market research, competitor analysis, planning)
- Monthly strategy update
- Basic dashboard

Tier 2 (Strategy + Execution): $1,499/month

- All of Tier 1, plus:
- SEO agent (content calendar, publishing)
- Ads agent (campaign setup, basic management)
- Outreach agent
- Monthly execution report

Tier 3 (Full Stack): $3,999/month

- All of Tier 2, plus:
- Lead qualification chatbot
- Sales agent
- Support agent
- Retention agent
- Daily/weekly automated reports
- Dedicated account manager

Tier 4 (Enterprise): Custom pricing

- Custom agents for your specific workflow
- Integration with your existing tools
- Dedicated Slack channel for support
- Quarterly business reviews

Option 2: Revenue Share

- Free to use, system takes 5–15% of incremental revenue attributed to AIMAS
- Only works if customers allow revenue tracking
- High alignment: system succeeds only if customer succeeds

Option 3: Agency Model

- White-label the system for digital agencies
- Agencies sell to their clients at 2–3x your cost
- License: $2,000–5,000/month per agency

---

8. Key Differentiators vs. Competitors

Feature	AIMAS	Existing Tools
Multi-vertical support	Pre-configured for 10+ verticals	Single-purpose (SEO-only, ads-only, etc.)
Auto-discovery	Autonomously finds competitors, trends	Requires manual input
Full funnel	Awareness to retention	Usually one stage (e.g., ads or support)
Consolidated metrics	Single dashboard across all channels	Siloed: GA, Ads Manager, email platform, CRM all separate
Real agent orchestration	Agents communicate, share context	Tools are independent
No setup needed	Plug in business info, AI builds strategy	Steep learning curve, manual setup
Autonomous execution	Publishes content, runs ads, sends emails	Requires approval steps, human intervention
Continuous optimization	Daily adjustments, weekly recommendations	Static reports, humans decide what to do

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9. Risks & Mitigations

Risk	Impact	Mitigation
Competitor backlash	Bad press if AI scraping identified	Use official APIs where possible, respect robots.txt
Data privacy	GDPR/CCPA violations	Explicit consent, data deletion, privacy audit
AI output quality	Bad content, spam emails hurt brand	Human approval gates for all customer-facing content
Over-automation	AI does something customer didn't want	Clear constraints, weekly recommendations for approval, not auto-execute by default
Negative ROI	System spends budget without return	Built-in safeguards (CAC thresholds, budget caps, performance monitoring)
Churn	Customers leave if not seeing results	90-day money-back guarantee, free strategy review, proactive support

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10. Success Metrics

Product:

- Time to first campaign launch: <7 days (user inputs info → system runs)
- % of users seeing positive ROI within 60 days: >70%
- Monthly churn rate: <5%
- NPS: >60

User Outcomes:

- Avg CAC reduction: 20–30%
- Avg LTV increase: 15–25%
- Avg time saved on marketing ops: 20–30 hours/month per user
- Revenue per $1 marketing spend: 5–10x

---

11. Roadmap (12-Month View)

Q1:

- MVP launch (strategy agents)
- Soft launch with 50 beta customers
- Vertical-specific presets (SaaS, E-commerce, Services)

Q2:

- Execution agents (SEO, Ads, Outreach)
- Chatbot integration
- Airtable + Zapier ecosystem

Q3:

- Sales + Support agents
- CRM integrations (HubSpot, Pipedrive)
- Real-time analytics dashboard

Q4:

- Optimization agents (testing, budget allocation)
- White-label enterprise version
- 500+ paying customers, $200k MRR target

Year 2:

- Expansion to 10+ integrations
- AI-powered content generation (in-house)
- Voice/audio agents (autonomous sales calls)
- Multi-language support

---

12. Technical Stack Summary

Core:

- **LLM**: Claude 3.5 Sonnet (reasoning) + o1 (data analysis)
- **Agentic Framework**: CrewAI or LangGraph
- **Backend**: Python + FastAPI
- **Database**: PostgreSQL (Supabase) + Pinecone (vector DB)
- **Job Queue**: Celery + Redis (async tasks)
- **Frontend**: React or Next.js (dashboard)

Integrations (15+ APIs):

- Google (Search Console, Analytics, Ads)
- Meta (Facebook Ads)
- Ahrefs, SEMrush
- WordPress, Webflow
- HubSpot, Pipedrive, Intercom
- Stripe, Lemlist, etc.

Deployment:

- AWS (EC2, RDS, Lambda) or Vercel (frontend)
- Docker containers for scalability
- GitHub Actions for CI/CD

Conclusion

AIMAS (Universal AI Marketing Automation System) is a comprehensive, modular AI framework that autonomously discovers market opportunities, orchestrates multi-channel campaigns, and optimizes for ROI—applicable to any business vertical.

Core value:

- **Save time:** 20–30 hours/month on marketing ops
- **Reduce CAC:** 20–30% through intelligent channel allocation and optimization
- **Increase LTV:** Retention and upsell automation
- **Focus:** Humans focus on strategy and creativity, AI handles execution and optimization

Go-to-market:

- SaaS subscription ($299–$3,999/month depending on tier)
- 90-day free trial for new customers
- White-label for agencies
- Revenue share option for high-growth users

First 12 months:

- Launch MVP with strategy agents
- Add execution agents
- Reach 500+ customers, $200k MRR
- Become the "operating system" for modern marketing teams